Abu Jafar Saifullah

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Research Interests

AI for Software Engineering (LLM coding agents), Software Accessibility, Program Analysis and Automated Repair

Education

University of Dhaka — Dhaka, Bangladesh
M.Sc. in Software Engineering (completed; final result pending); ***GPA: 3.92/4.00*** — *Apr 2025 – Sep 2026*
B.Sc. in Software Engineering; ***GPA: 3.89/4.00*** — *Jan 2019 – Dec 2023*

Notre Dame College — Dhaka, Bangladesh
Higher Secondary Certificate; ***GPA: 5.00/5.00*** — *Jun 2016 – Jun 2018*

Publications

Conference Papers

[C1] **Abu Jafar Saifullah**, Tasmia Zerin, Zerina Begum, Kazi Sakib. "NavA11y: A Dynamic Analysis Approach for WCAG 2.4 Focus-Behavior Evaluation." **ENASE 2026**, full paper, pp. 1019–1030. [Paper] [Code]

Under Review

[U1] Syed Fatiul Huq, Md. Asif Haider, **Abu Jafar Saifullah**, Thomas Zimmermann, Sam Malek. "Vibing Accessibly: Investigating the Accessibility of Websites Generated by LLM Coding Agents." 2026.

Research Experience

DSSE Research Group, IIT, University of Dhaka — Dhaka, Bangladesh
Graduate Researcher; supervisor Dr. Zerina Begum, co-supervisor Dr. Kazi Sakib — *Apr 2025 – Sep 2026*

- M.Sc. thesis (completed): *Detecting and Repairing Keyboard Focus-Behavior Violations through Dynamic Analysis.* Built **NavA11y** to detect these violations and **RepairA11y** to fix them.
- NavA11y drives a real browser through each page's keyboard navigation and checks every focus stop: **100% precision and recall** on a 22-page labeled dataset; **2,947 violations** on 26 production websites at 90% precision, confirmed by two independent raters.
- Built and released the 22-page labeled dataset (extending a 6-page public test suite), the first to cover all five WCAG 2.4 focus-behavior criteria.

SEMERU Lab, William & Mary — Remote
Research Intern, supervised by Dr. Denys Poshyvanyk — *Aug 2023 – Dec 2023*

- Built a programming-language-specific tokenizer to improve NLP model predictions on software engineering tasks.

Research Projects

RepairA11y | *Node.js, Playwright* | *M.Sc. Thesis, 2026*

- Fixes the violations NavA11y finds: generates a CSS or attribute fix with rules or an LLM, applies it in a live browser, and re-checks the page. On 32 test cases, rules fixed **29** and the LLM **25**, with no new violations introduced.

Clippy | *Angular, Node.js, AWS S3* | GitHub

- PDF reader for researchers with knowledge-graph extraction, citation-network visualization, and paper recommendation.

BPE Tokenizer | *Python, Flask* | PyPI

- Byte Pair Encoding tokenizer built from scratch and published on PyPI, with a web interface for training on custom text; groundwork for the SEMERU Lab tokenizer.

Industry Experience

Optimizely, Content Marketing Platform — Dhaka, Bangladesh

Software Engineer II (promoted from SE I, Apr 2026) — *Apr 2024 – Present, Full-time*

- Integrated an internal AI SDK into the product end to end, including its service, release pipeline, and usage tracking across **10+ product surfaces**.
- Built the product's AI document editor: page context for LLM prompts, interactive form blocks, and rendering of AI-generated content.
- Led the migration to the public **Axiom** design system (**26 merged upstream PRs**; migration guides adopted by other teams) and brought Dark Mode to **15+ surfaces**.

Orion Informatics Ltd. — Dhaka, Bangladesh

Software Engineer — *Jan 2023 – Jan 2024, Full-time*

- Built reporting and data-visualization features for an attendance management system, and a cloud webhook service for trading alerts (ASP.NET Core, Azure).

Other Projects

Blueberry Auto | *React, Hono, Supabase* | Live | *Lead Developer, 2025 – Present*

- Vehicle auction and marketplace platform; leading the end-to-end build as the contracted developer (**150+ merged PRs**).

API Miners GPT | *React, ASP.NET Core, OpenAI API* | GitHub | *BUET Hackathon 2023*

- Chat-with-PDF assistant built on GPT, with voice input, OCR for images, and image generation.

EmitXplorer | *React, ASP.NET Core, TensorFlow* | GitHub | *NASA Space Apps 2023*

- Game-style visualization of satellite greenhouse-gas data, with an ML model that predicts CO_2 concentration.

Mr. Bin | *MERN, Spring Boot 3, TensorFlow* | GitHub | *JavaFest 2023*

- Recycling and resale marketplace that classifies listed items from photos with an ML model.

Technical Skills

Research Methods: Dynamic Program Analysis, Mining Software Repositories (SZZ), Statistical Evaluation (Kendall τ, Cohen's κ, PABAK), Inter-rater Agreement

Languages: TypeScript, JavaScript, Python, Java, C#, C++

Tools & Frameworks: Playwright, axe-core, Selenium, HuggingFace Transformers, scikit-learn, pandas, React, Node.js

Infrastructure: AWS, Azure, Docker, GitHub Actions, PostgreSQL, MongoDB

Awards & Achievements

ICT Division Fellowship, Government of Bangladesh: research funding for M.Sc. thesis (2026)

Winner, Optimizely Global Hackathon 2025: company-wide; our team built Spotlight search, later shipped to production

Top 6, ICSE SCORE 2023 (international student contest on software engineering), for Clippy

Best UI/UX, BUET Hackathon 2023, for API Miners GPT

Talks & Service

Talk: "NavA11y: A Dynamic Analysis Approach for WCAG 2.4 Focus-Behavior Evaluation," ENASE 2026 (presented remotely), May 2026

Internal Reviewer, DSSE Research Group (2025–2026): reviewed group members' paper drafts before submission

References

Dr. Kazi Sakib
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Dr. Zerina Begum
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